

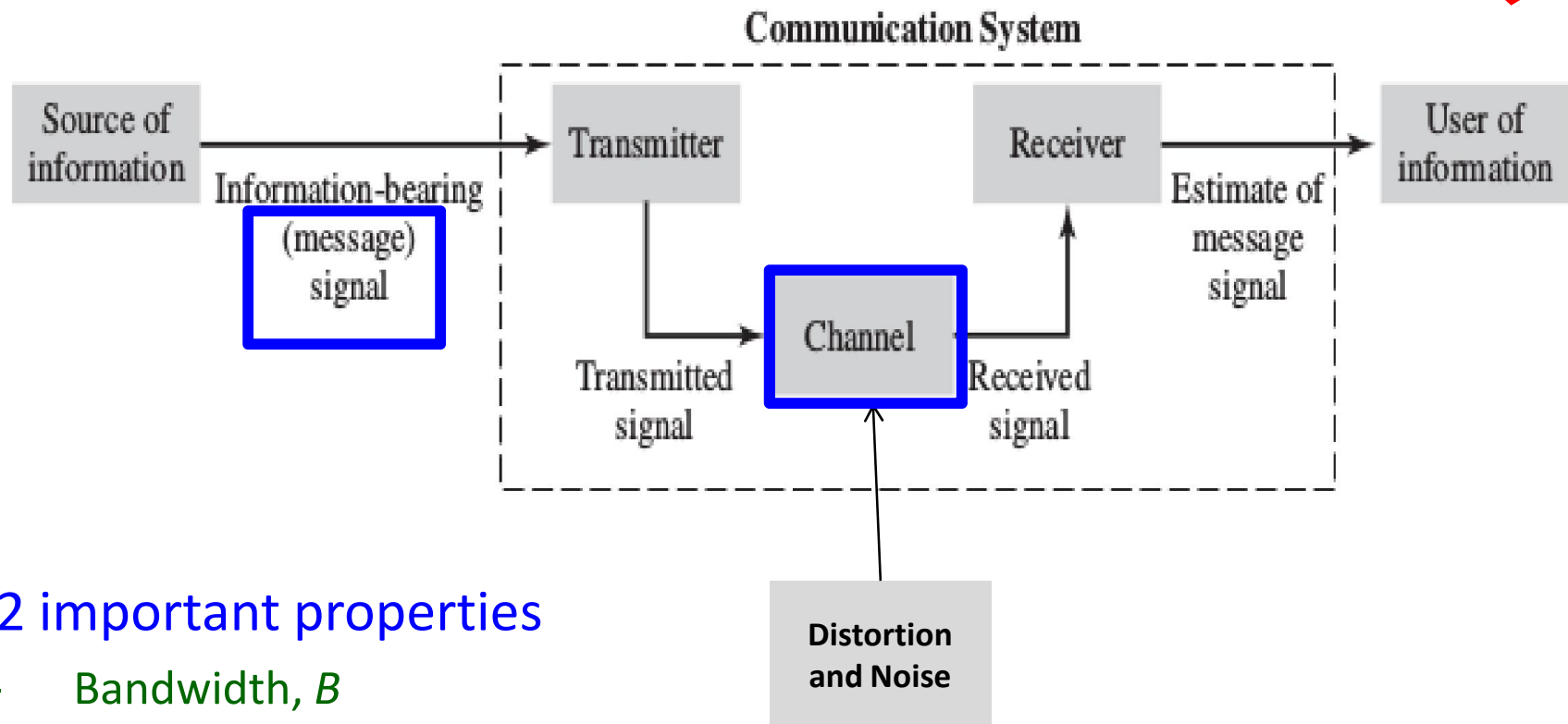
CSE 311:

Data Communication

Instructor:
Dr. Md. Monirul Islam

Channel & signal Characteristics

Review

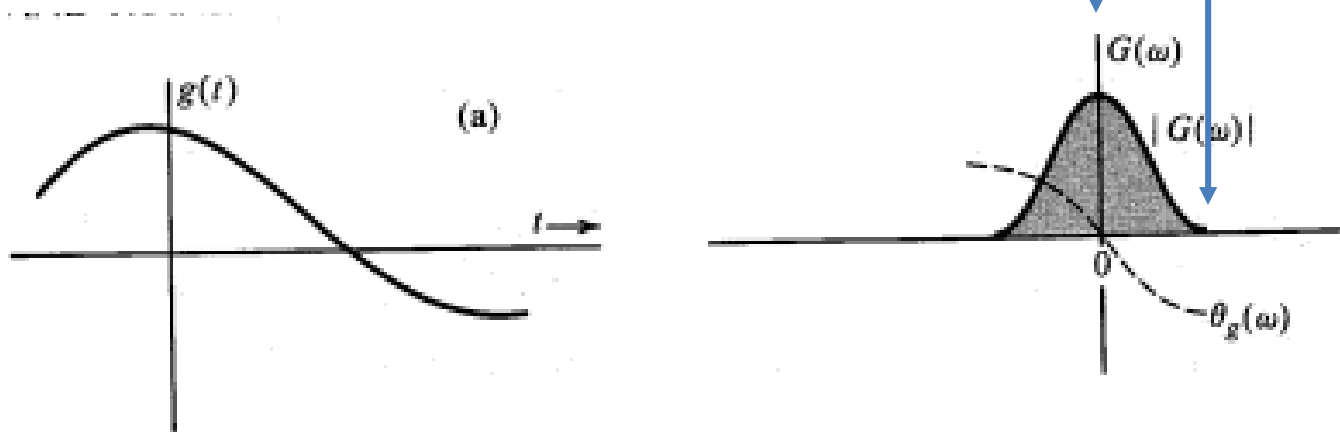


- 2 important properties
 - Bandwidth, B
 - Signal power, P_s

Channel & signal Characteristics

Review

- Bandwidth
 - Channel Bandwidth
 - Range of frequencies that a channel can transmit
 - Ex: if transmit 0-5 kHz frequencies, Channel bandwidth $B = 5$ kHz
 - Signal bandwidth
 - Maximum frequency that is available in a signal

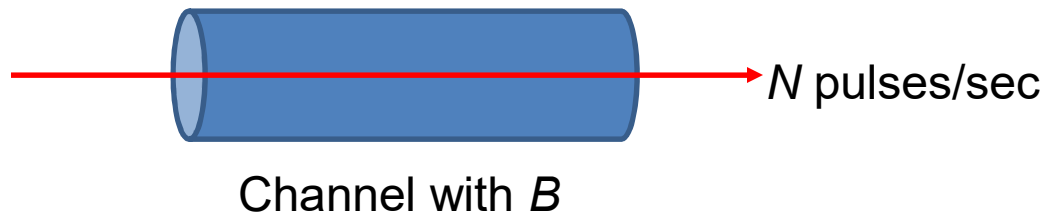


Channel & signal Characteristics

- More transmission speed requires channel with higher bandwidth

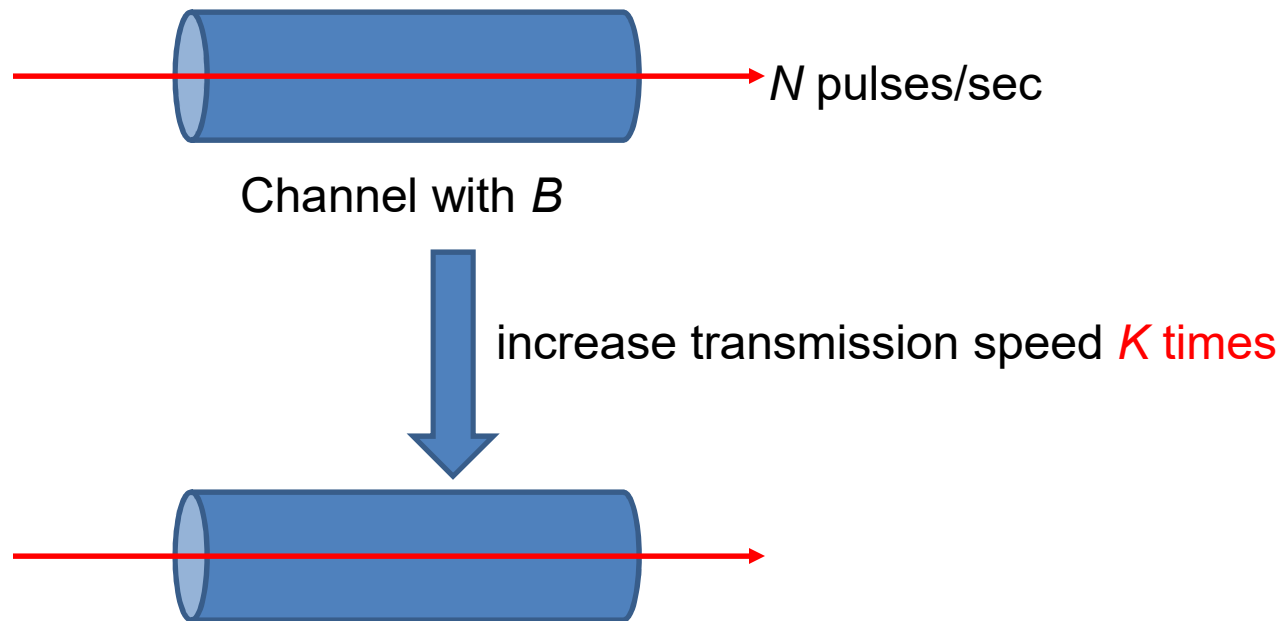
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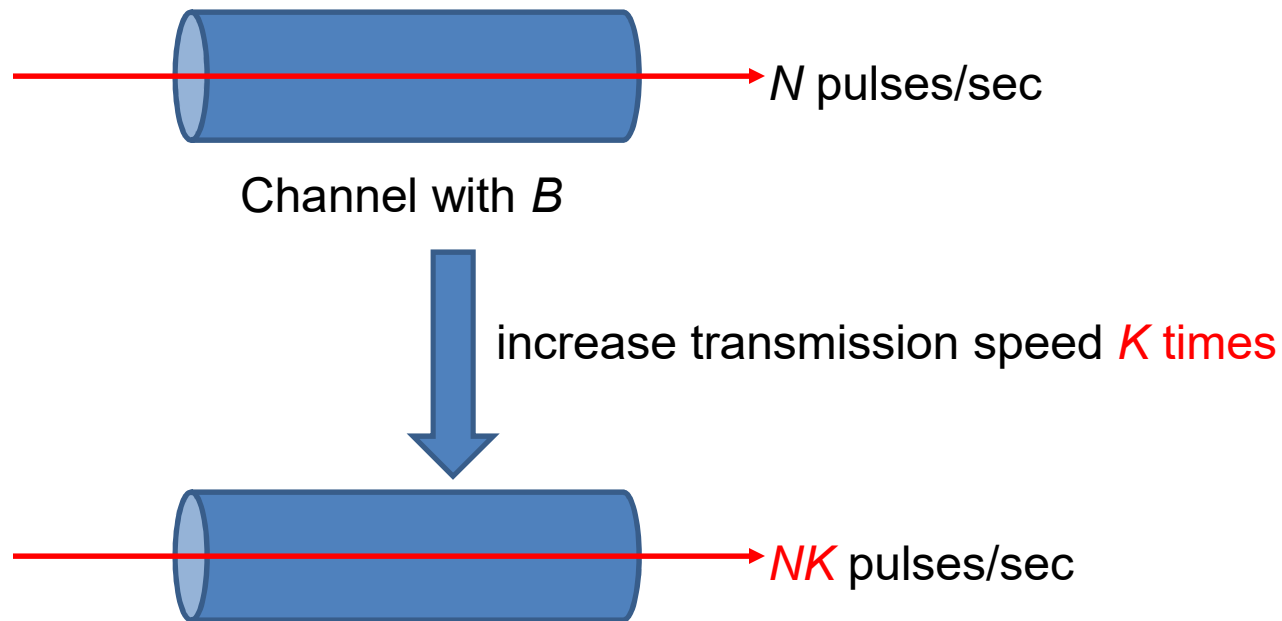
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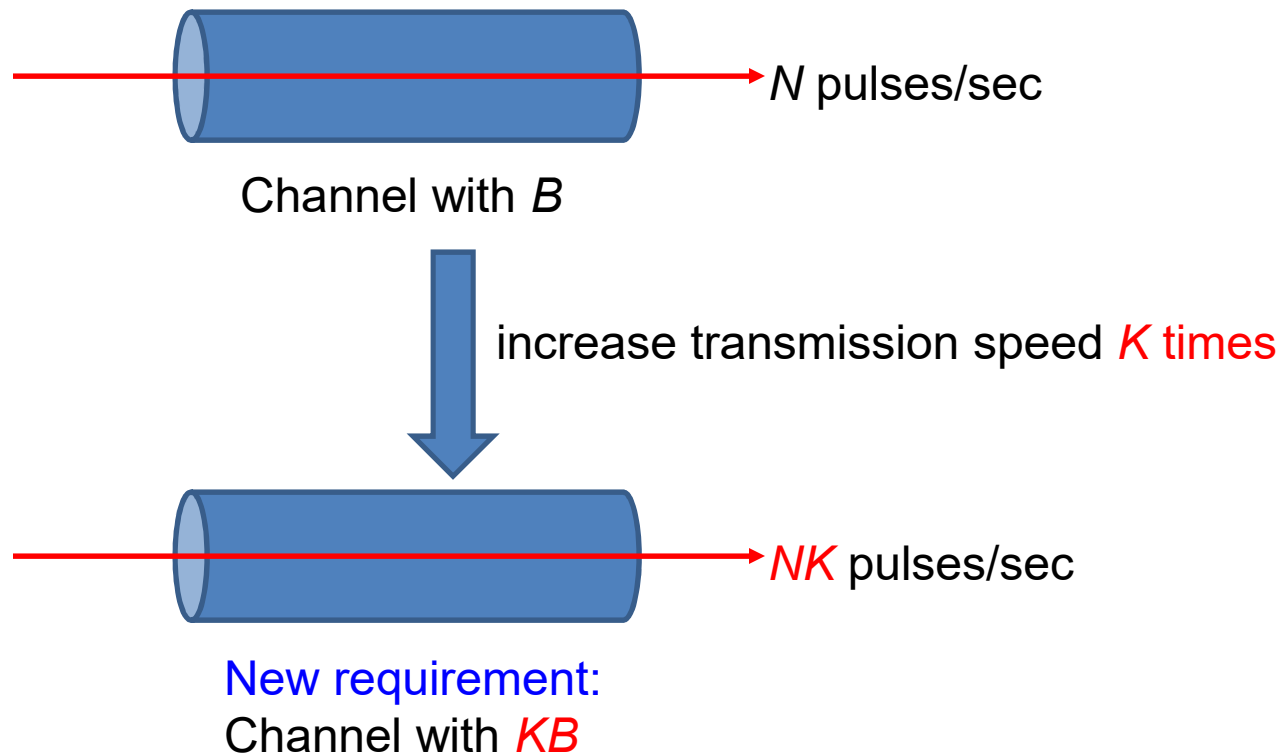
Channel & signal Characteristics

- More transmission speed requires channel with higher bandwidth



Channel & signal Characteristics

- More transmission speed requires channel with higher bandwidth



Channel & signal Characteristics

- Signal Power, P_s
 - Dual role
 - Higher Quality
 - Less channel bandwidth

Channel & signal Characteristics

- Signal Power, P_s for higher quality
 - $P_s++ \rightarrow SNR++$
 - Can maintain minimum SNR for longer distance
 - Higher SNR means
 - more noise immunity
 - Easier detection of pulses

Channel & signal Characteristics

- Signal Power, P_s needs less channel bandwidth
 - Higher signal P_s eases the channel

Channel & signal Characteristics

- Signal Power, P_s needs less channel bandwidth
 - Higher signal P_s eases the channel
 - Shannon's limit on channel capacity

$$C = B \log_2(1 + SNR) \text{ bit/s}$$

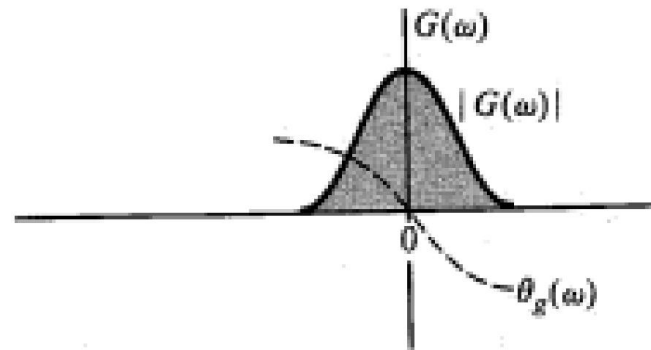
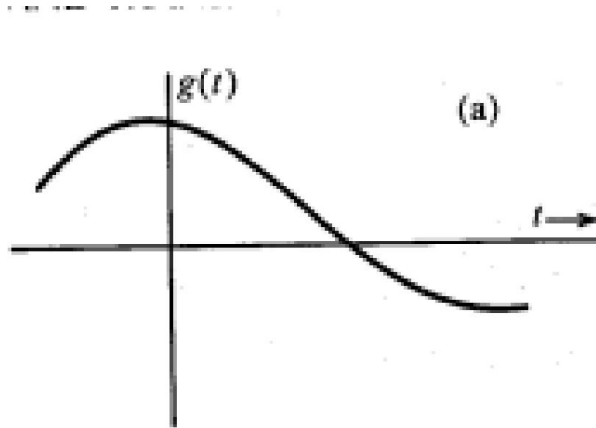
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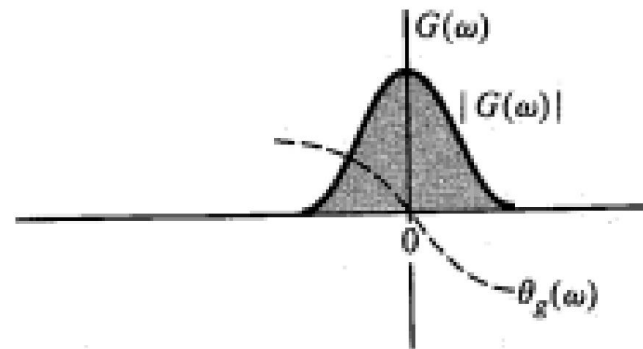
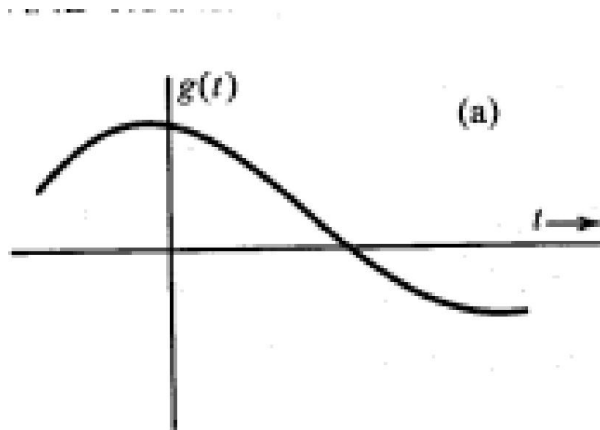
What happens if $SNR = \infty$?

Channel & signal Characteristics

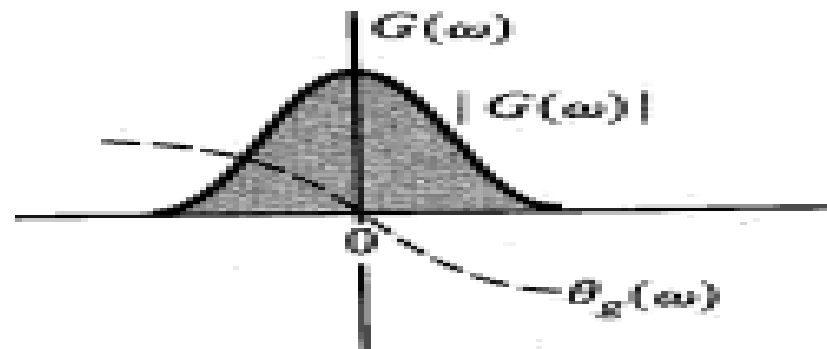
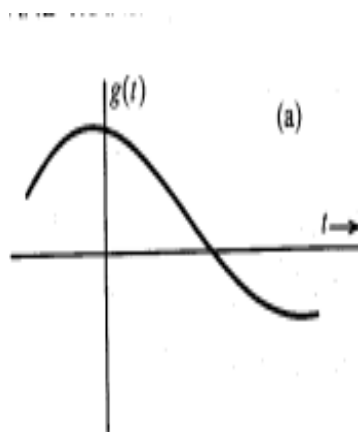


Baseband signal

Channel & signal Characteristics



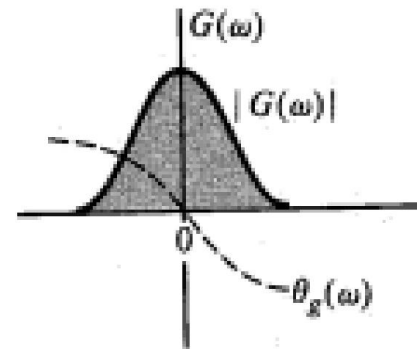
Baseband signal 1



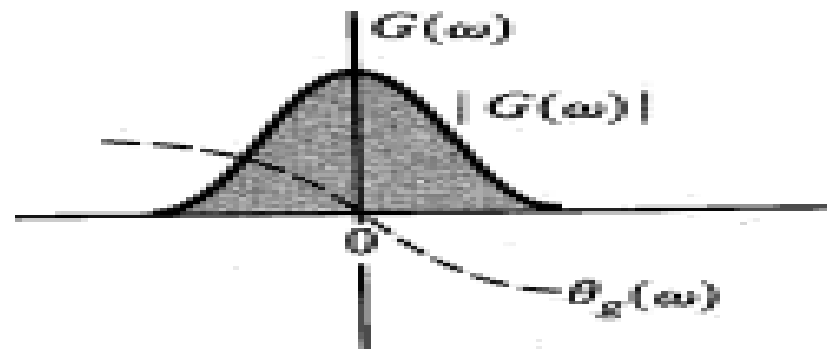
Baseband signal 2

Channel & signal Characteristics

- Assume, these 2 signals need to be transmitted



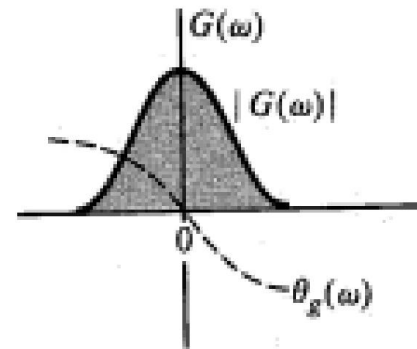
Baseband signal 1



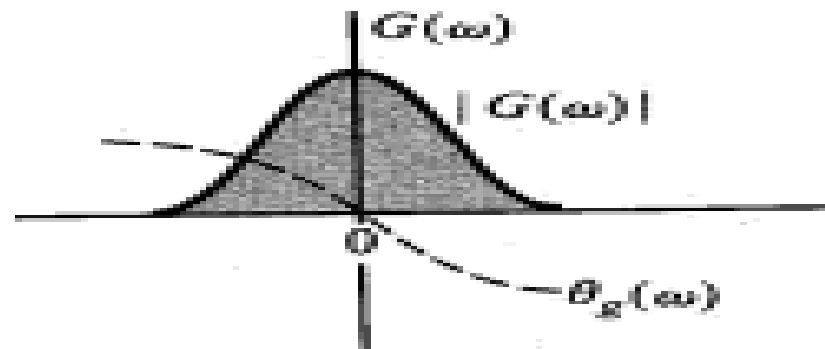
Baseband signal 2

Channel & signal Characteristics

- Assume, these 2 signals need to be transmitted
 - 2 different channels?



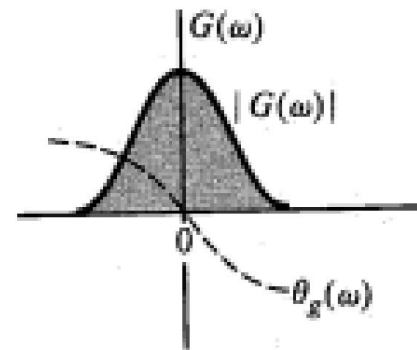
Baseband signal 1



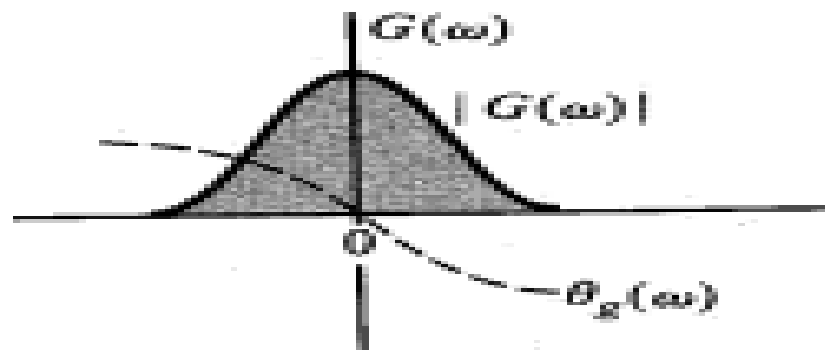
Baseband signal 2

Channel & signal Characteristics

- Assume, these 2 signals need to be transmitted
 - Even, channel bandwidth and signal bandwidth may NOT match!!



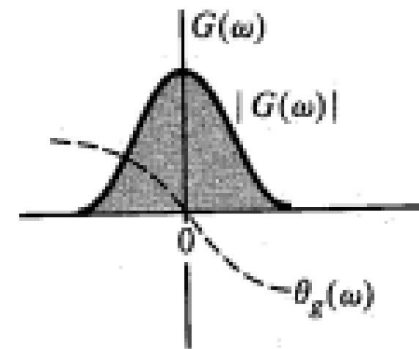
Baseband signal 1



Baseband signal 2

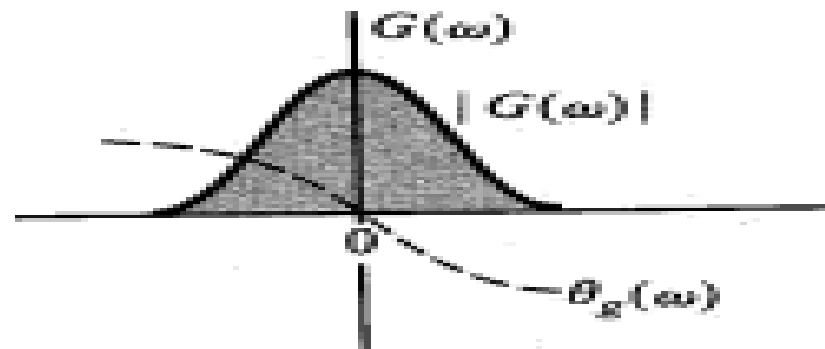
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Baseband signal 1

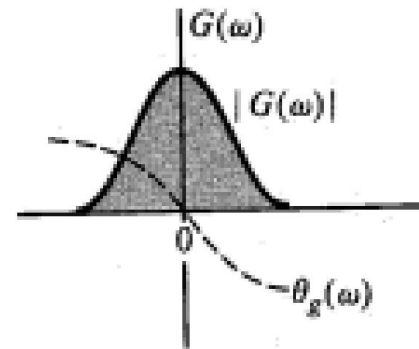
Solution is
modulation



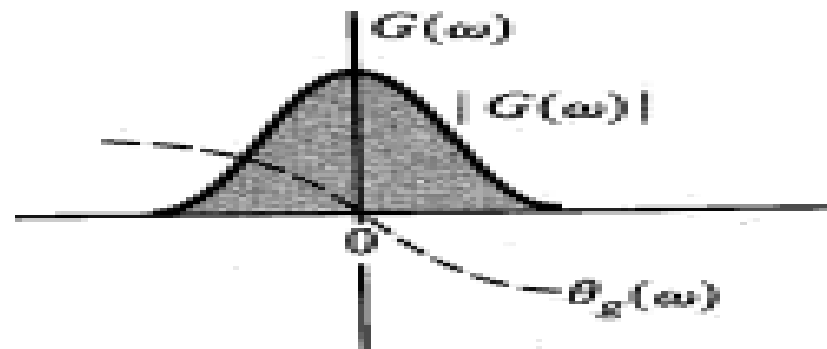
Baseband signal 2

Channel & signal Characteristics: *modulation*

- Use carrier signal to shift these 2 signals in different frequency positions



Baseband signal 1

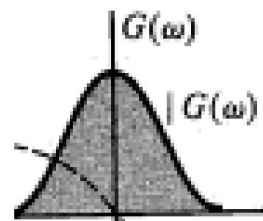


Baseband signal 2

Channel & signal

Characteristics: *modulation*

- Use carrier signal to shift these 2 signals in different frequency positions

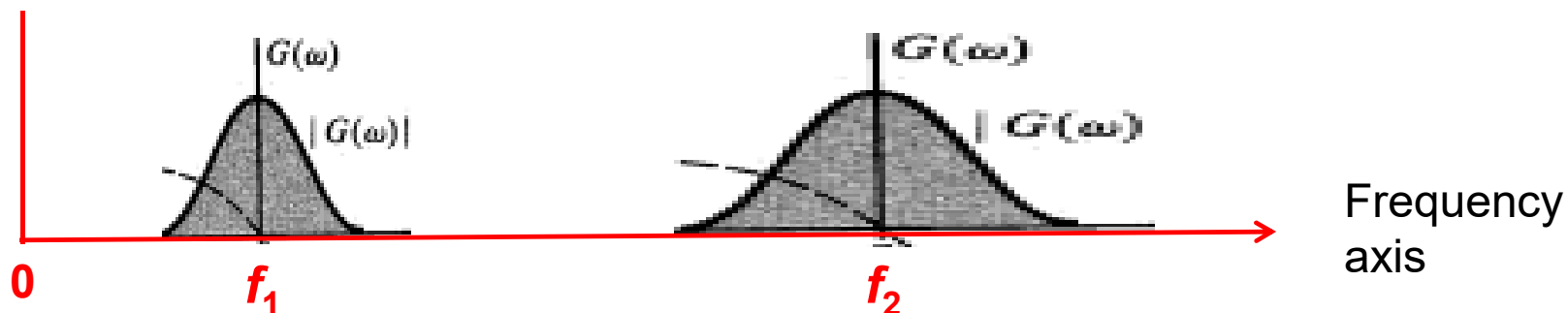


Baseband
signal 1



Baseband
signal 2

After
modulation:

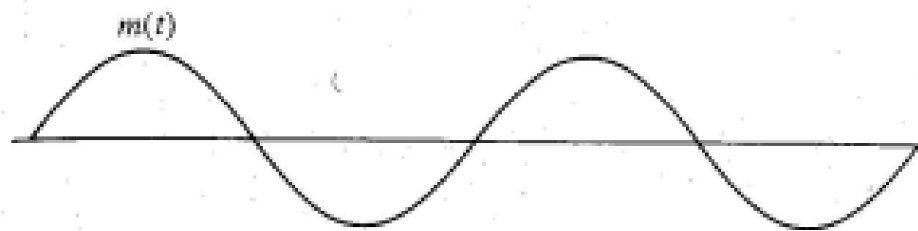


Channel & signal

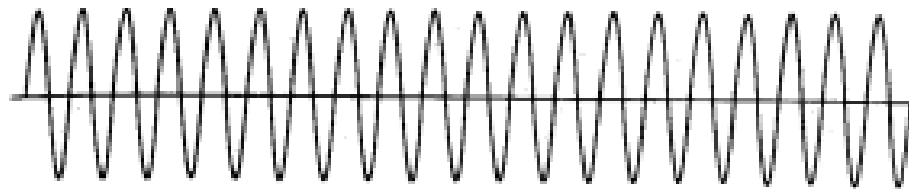
Characteristics: *modulation*

- other examples of modulation
(shown in time domain)

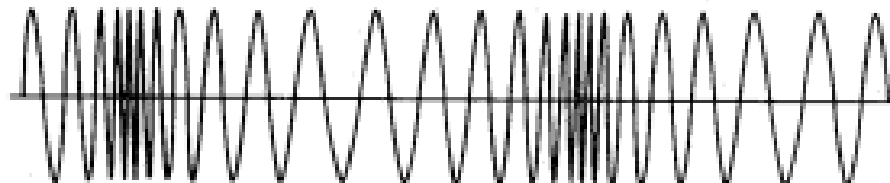
Baseband
signal



Carrier signal



Modulated
signal

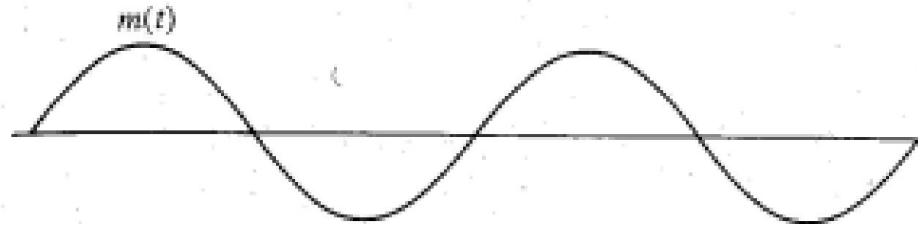


Channel & signal

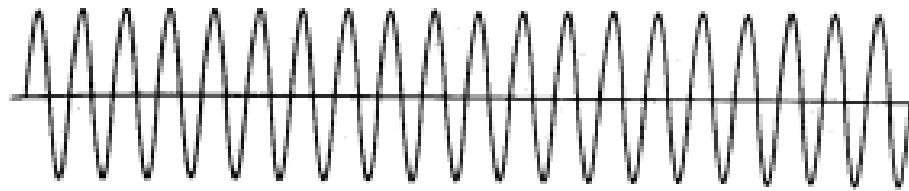
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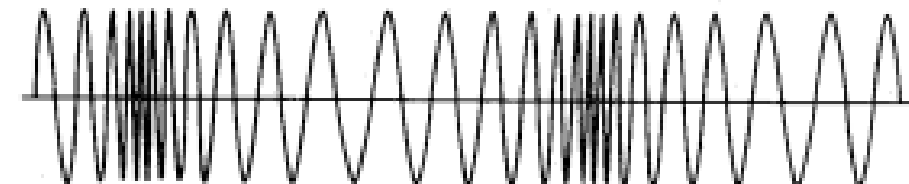
Baseband
signal



Carrier signal



Modulated
signal



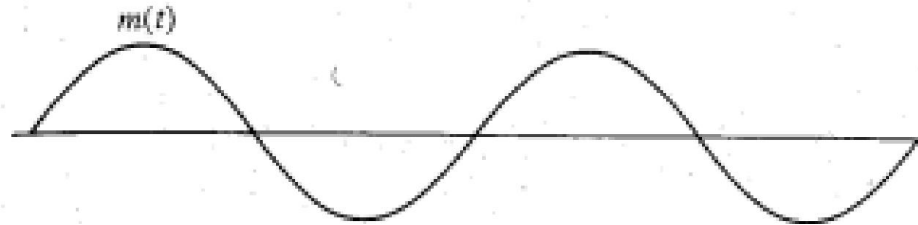
**Frequency
modulation
(FM)**

Channel & signal

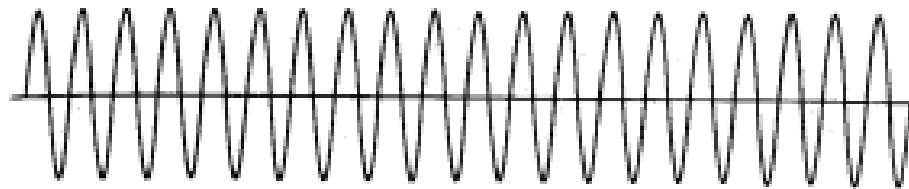
Characteristics: *modulation*

- other examples of modulation
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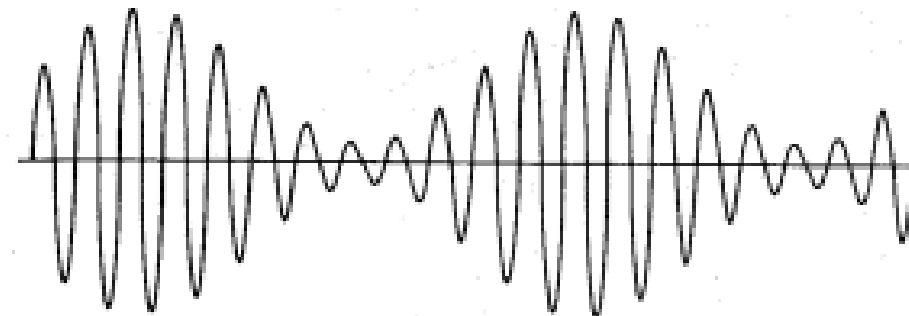
Baseband
signal



Carrier signal



Modulated
signal



**Amplitude
modulation
(AM)**

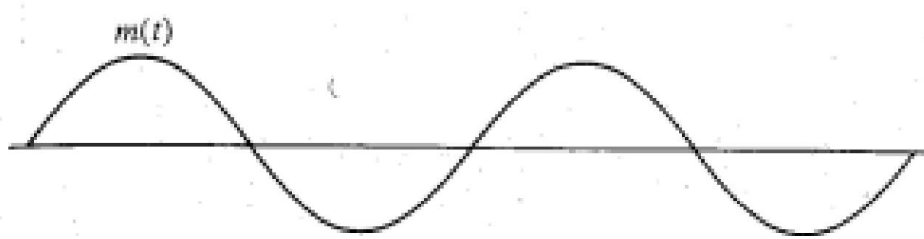
Channel & signal

Characteristics: *modulation*

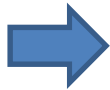
- Phase modulation (PM)
- Changes phase angle of the signal

Channel & signal Characteristics: *modulation*

Other reasons for modulation



Low frequency



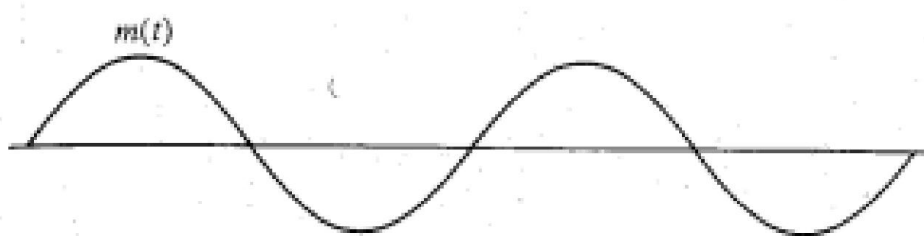
Higher
wavelength



Generating
antenna requires
higher dimension

Channel & signal Characteristics: *modulation*

Other reasons for modulation

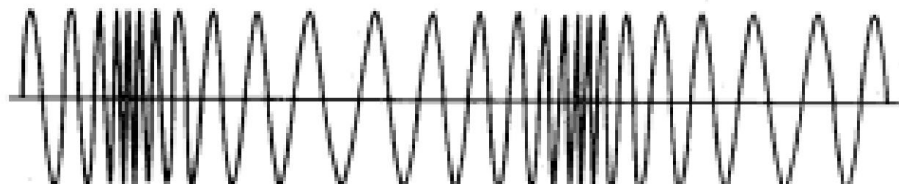


Low frequency → Higher wavelength → Generating antenna requires higher dimension

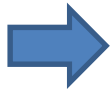
- Speech signal characteristics
 - Frequency range: 100 – 3000 Hz
 - Wavelength 100 to 3000 km
 - requires *impractically* large antenna

Channel & signal Characteristics: *modulation*

Other reasons for modulation



Modulating a High
frequency carrier



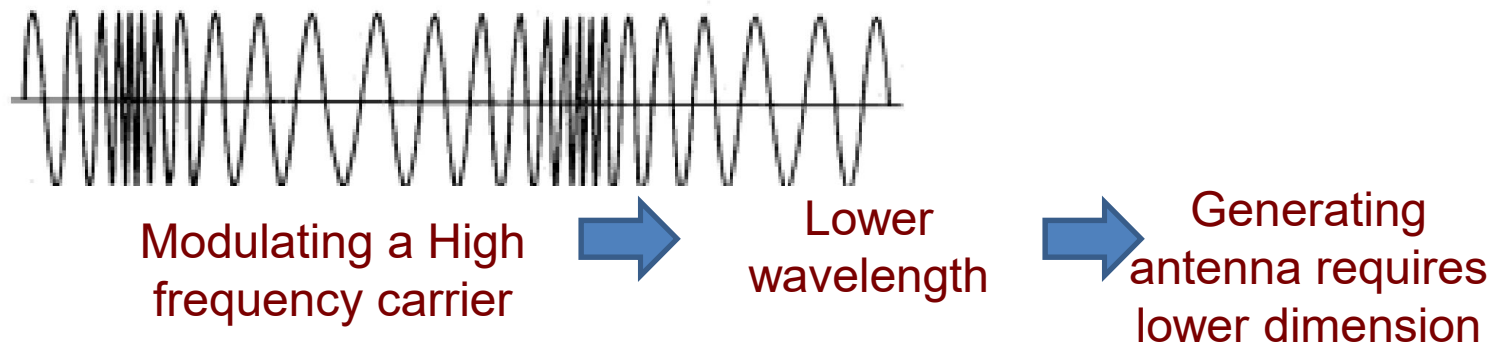
Lower
wavelength



Generating
antenna requires
lower dimension

Channel & signal Characteristics: *modulation*

Other reasons for modulation

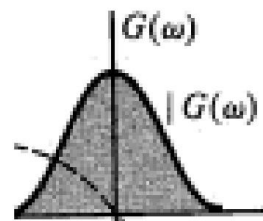


- 10 MHz carrier signal characteristics
 - Wavelength 30 m
 - requires antenna of size 3m

Channel & signal Characteristics: *modulation*

- Use carrier signal to shift these
2 signals in different frequency
positions

recall this example!

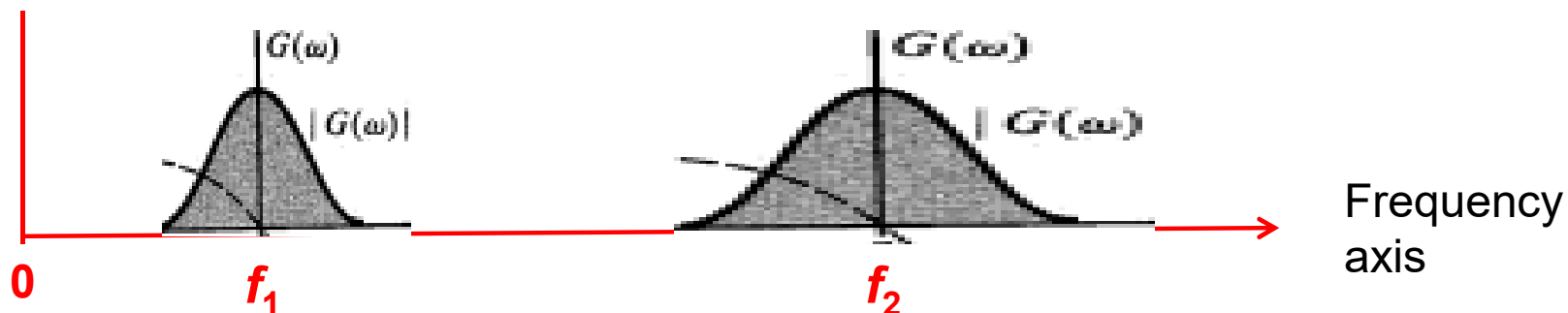


Baseband
signal 1



Baseband
signal 2

After
modulation:

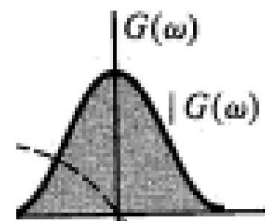


Channel & signal Characteristics: *modulation*

- Use carrier signal to shift these
2 signals in different frequency
positions

also called
*Frequency division
multiplexing (FDM)*

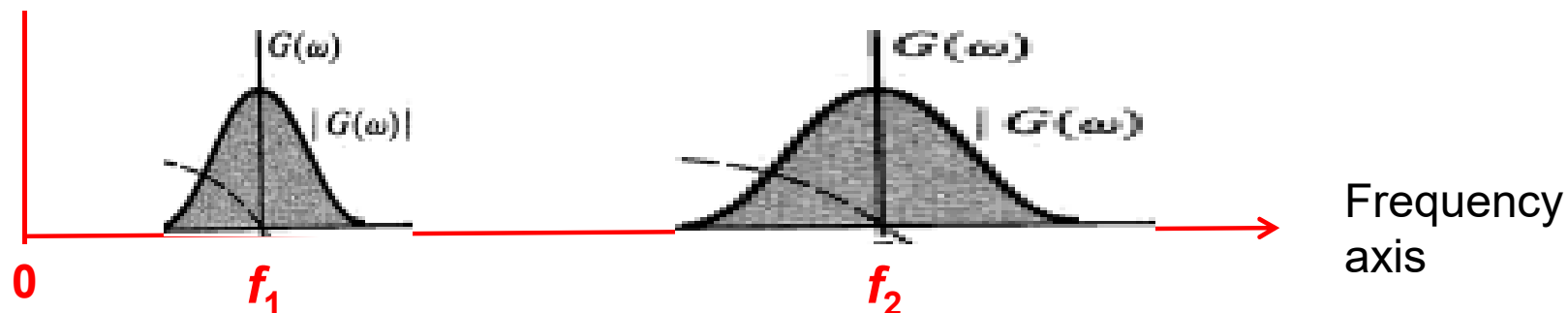
After
modulation:



Baseband
signal 1



Baseband
signal 2

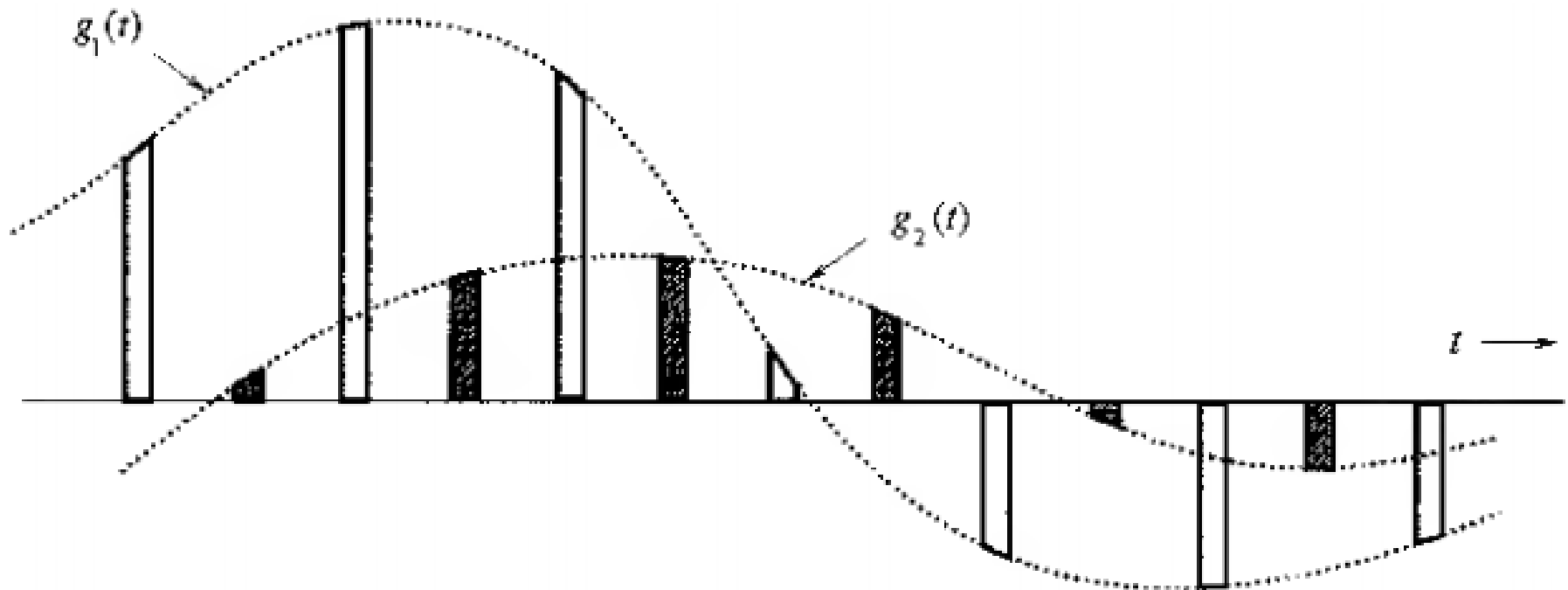


Channel & signal Characteristics: *modulation*

- Time division multiplexing (TDM)
 - Interleave pulses from different signals in time domain signal

Channel & signal Characteristics: *modulation*

- Time division multiplexing (TDM)
 - Interleave pulses from different signals in time domain signal



Channel & signal

Characteristics: *DeModulation*

- Done at the receiving end
 - Bandpass filter separates appropriate signal
 - Makes necessary corrections for amplitude, frequency and phase changes

Signal Characteristics: *digital source coding and error correction coding*

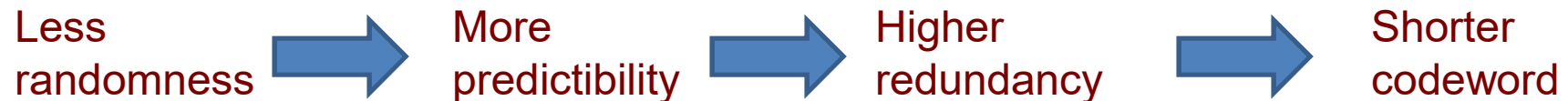
- 2 opposite procedures
- Digital source coding
 - An aggressive measure to **reduce redundancy**
- Error correction coding
 - A systematic **addition of redundancy** to detect/correct errors

Digital source Coding

- Removes redundancy
- Uses bandwidth as little as possible
- Related to randomness/predictability in data

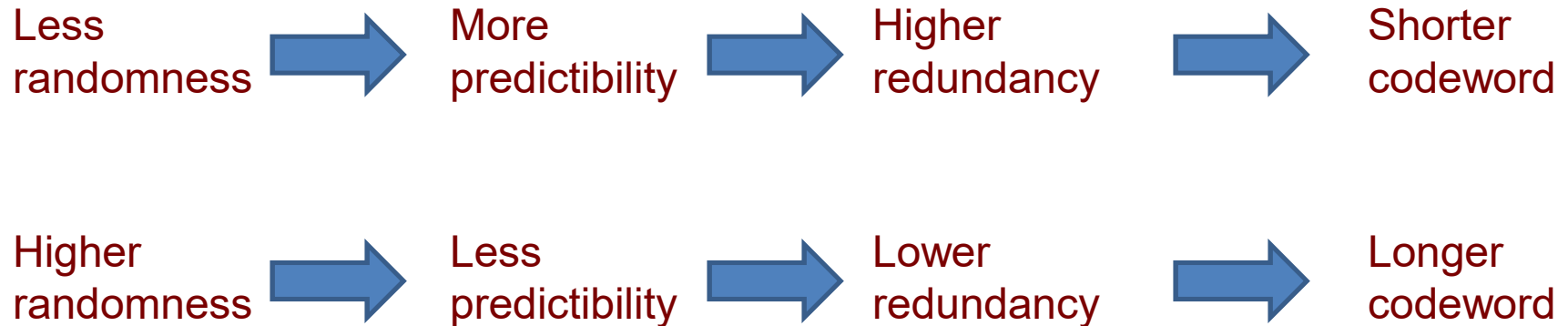
Digital source Coding

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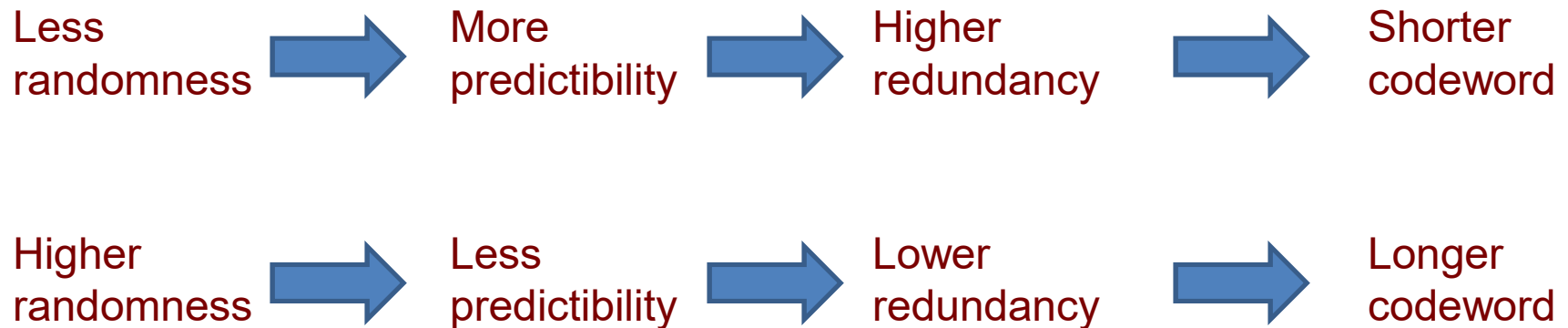
Digital source Coding

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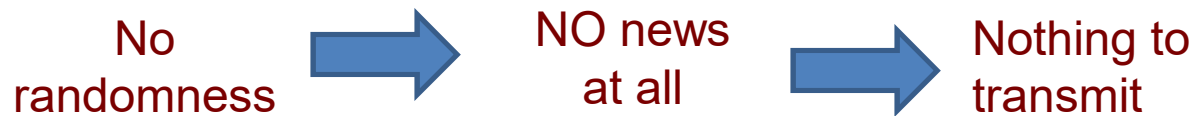


Digital source Coding

- Example: Morse code
 - Frequently occurring letters e, t, a: shorter codes
 - Less frequent letters: longer codes

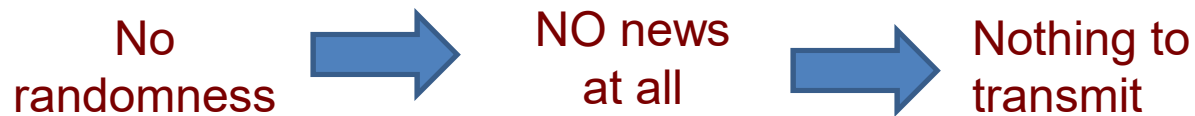


Digital source Coding



- A message with probability p ,
its *randomness or entropy* $= \log (1/p) = -\log (p)$

Digital source Coding



- A message with probability p ,
its *randomness or entropy* $= \log (1/p) = -\log (p)$

What happens when $p = 0$ and $p = 1$?

Error correction coding

- Add systemic redundancy
- 50% of English text is redundant
- Difficult to recover if error occurs in reduced text

Error correction coding

- Add systemic redundancy
- 50% of English text is redundant
- Difficult to recover if error occurs
- Error Detection: adding parity bit

Error correction coding

- Add systemic redundancy
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Assume to transmit **0001**

Error correction coding

- Add systemic redundancy
- 50% of English text is redundant
- Difficult to recover if error occurs
- Error Detection: adding parity bit

0001**1**



Add this bit to make
even no. of 1's

Error correction coding

- Add systemic redundancy
- 50% of English text is redundant
- Difficult to recover if error occurs
- Error Detection: adding parity bit

0001**1**



Add this bit to make
even no. of 1's

Detects single bit error; however
cannot detect even no. of errors or
cannot locate or correct single bit
error

Signals and systems

Topics

- Definition, classification and properties of signals
- Examples of some useful signals
- Fourier Series of Periodic Signals

Messages/Signals: Definition

Review

- A signal is a set of information or data.
- A signal is a function of independent variables that carry some information.
- A signal is a physical quantity that varies with time, space or any other independent variable by which information can be conveyed.

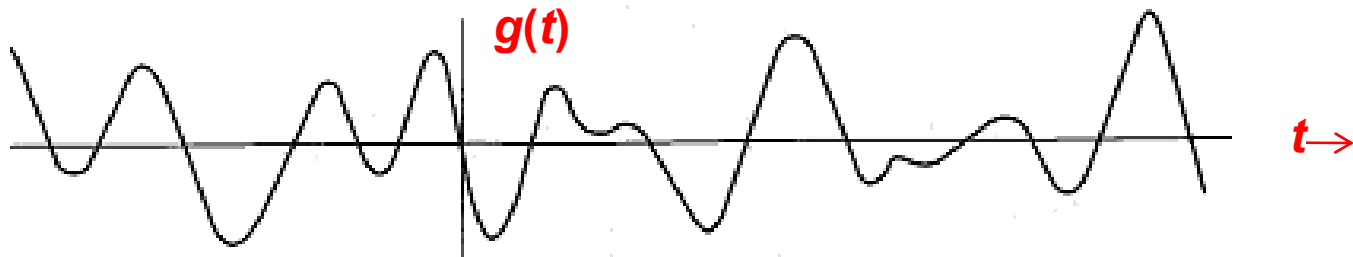
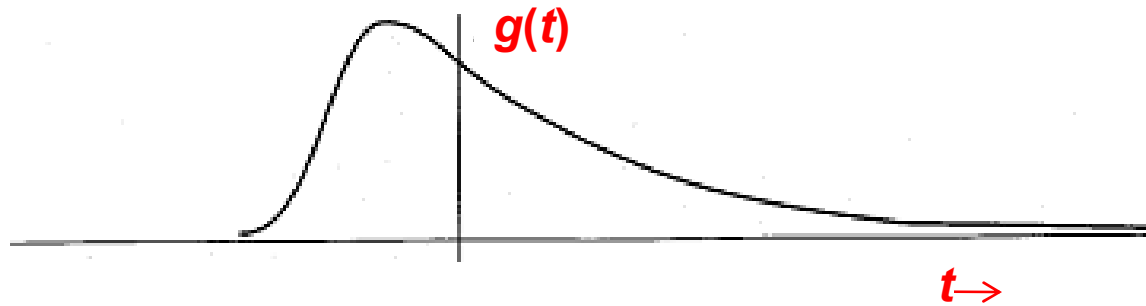
Example of Signals

Review

- Voice signal
- Telephone or television signal
- Monthly sales figure
- Opening or closing stock prices
- Charge density over a surface
- In this course we deal with signals that are functions of time.

Signal representation: Time Domain

Review



Systems

- Processes signals to
 - modify signal
 - extract information
- Example:
 - electrical, mechanical, hydraulic
 - USG, Telephone, Radar or television signal

Strength of a Signal, $g(t)$

- $g(t)$ is the voltage across a one ohm resistor
- Two quantities of strength
 - Signal energy, E_g
 - Signal power, P_g

Signal Energy, E_g

- E_g is the energy that voltage $g(t)$ dissipates on 1Ω resistor

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Mathematically,

$$E_g = \int_{-\infty}^{\infty} g^2(t) dt$$

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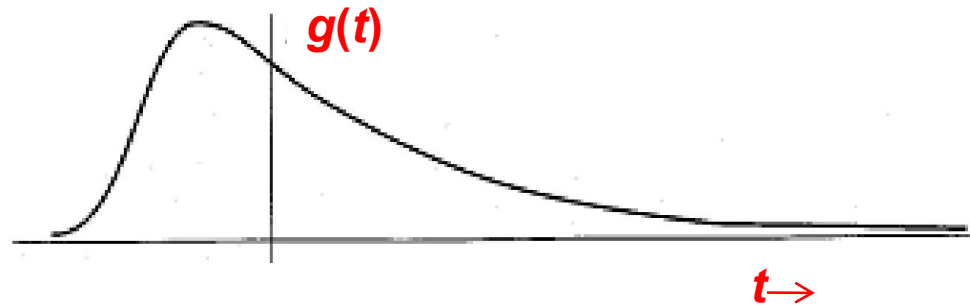
For complex valued signal,

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt$$

Signal Energy, E_g

$$E_g = \int_{-\infty}^{\infty} g^2(t) dt$$

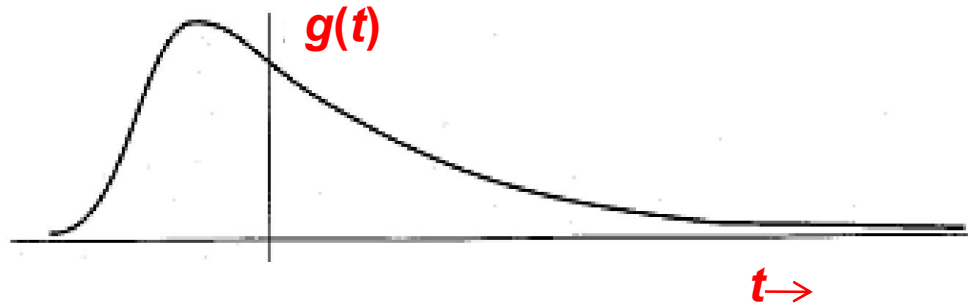
Consider this signal,



Signal Energy, E_g

$$E_g = \int_{-\infty}^{\infty} g^2(t) dt$$

Consider this signal,

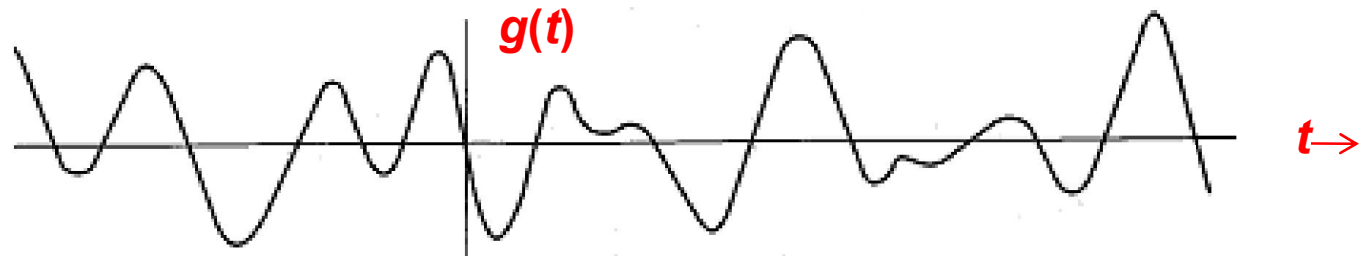


- $g(t) \rightarrow 0$ as $|t| \rightarrow \infty$
- E_g converges, is integrable, and has limit

Signal Energy, E_g

$$E_g = \int_{-\infty}^{\infty} g^2(t) dt$$

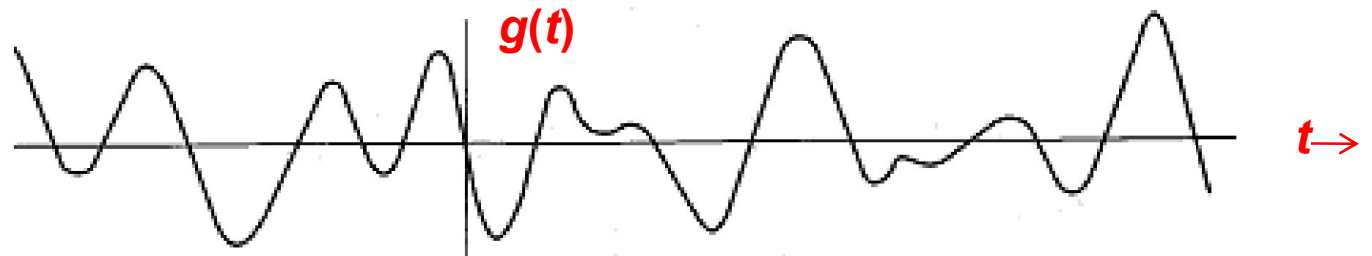
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Signal Energy, E_g

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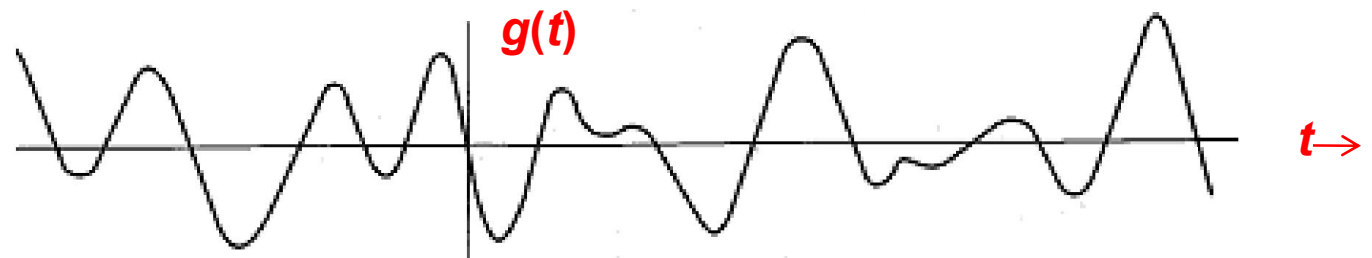
Consider this signal,



- As $t \rightarrow \infty$, amplitude of $g(t)$ **does NOT** go to zero
- E_g **does NOT converge**, is **NOT integrable**, and has **NO limit**

Signal Power, P_g

Consider this signal,



Strength is
measured using P_g

Signal Power, P_g

For real valued signal,

$$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g^2(t) dt$$

For complex valued signal,

$$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |g(t)|^2 dt$$

Signal Power, P_g

$$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g^2(t) dt$$

Real valued signal

$$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |g(t)|^2 dt$$

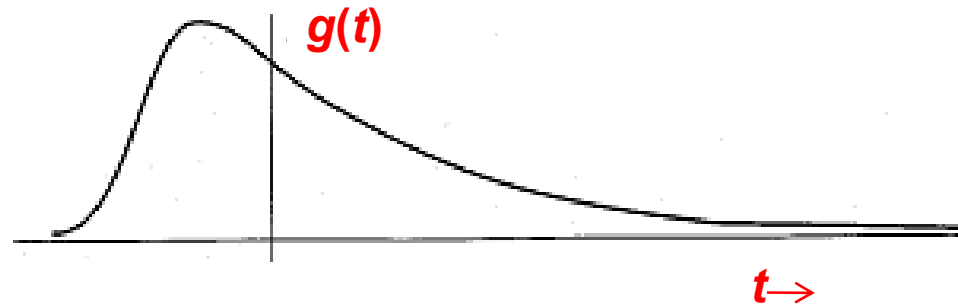
Complex valued signal

- P_g is the **time average** of signal amplitude square
- Also known as **mean square** of $g(t)$
- rms of $g(t) = \sqrt{P_g}$

Signal Power, P_g

$$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g^2(t) dt$$

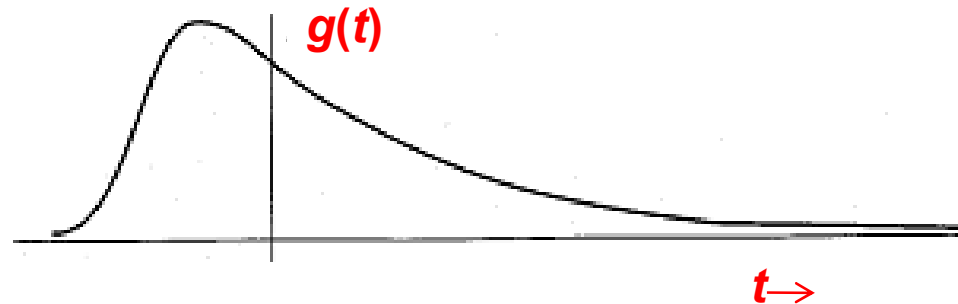
What is P_g of this signal?



Signal Power, P_g

$$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g^2(t) dt$$

What is P_g of this signal?

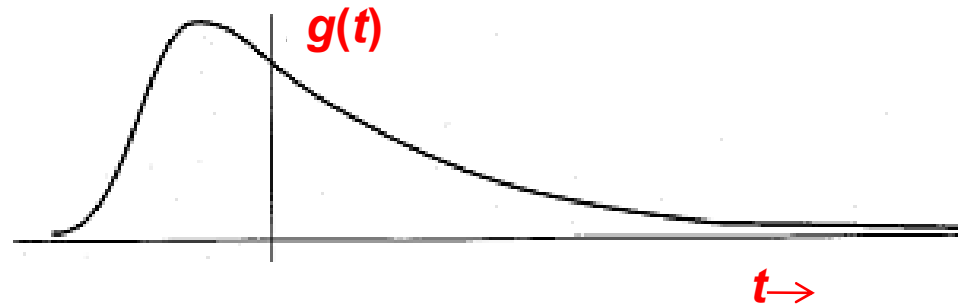


$$P_g = 0$$

Signal Power, P_g

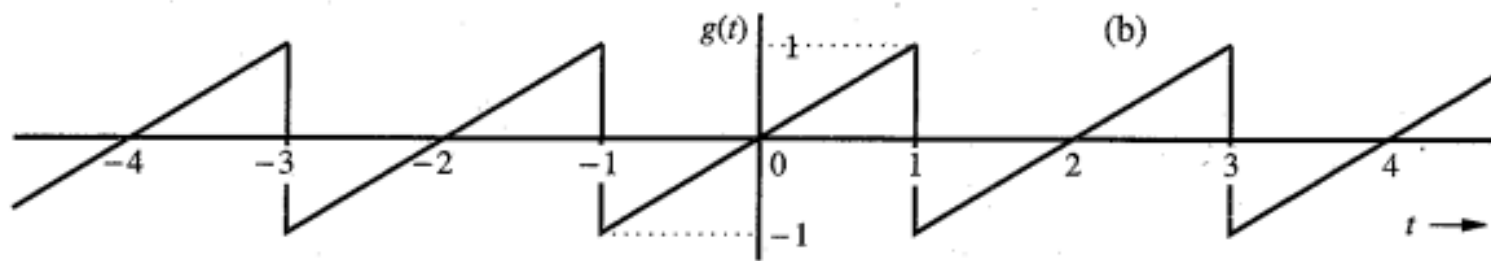
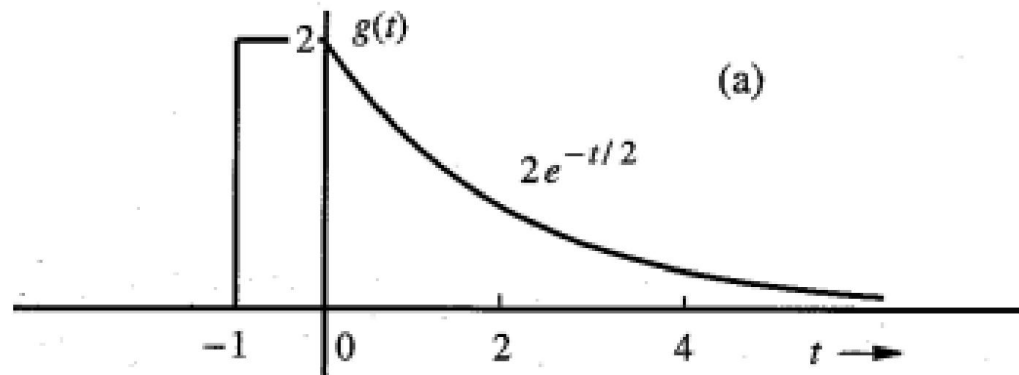
$$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g^2(t) dt$$

What is P_g of this signal?

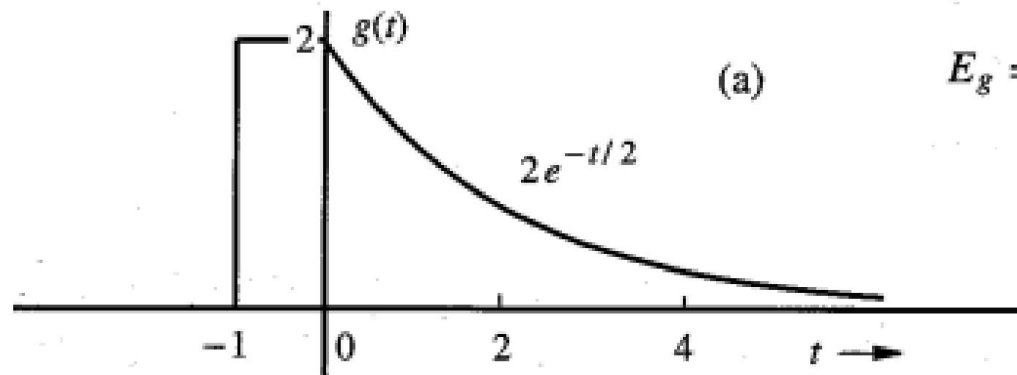


- P_g exists when the signal
- is periodic, or
 - has statistical regularity

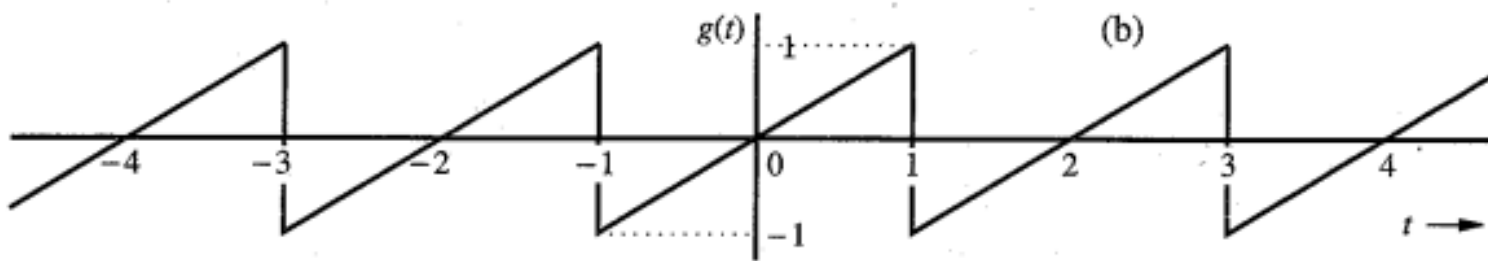
Find P_g or E_g



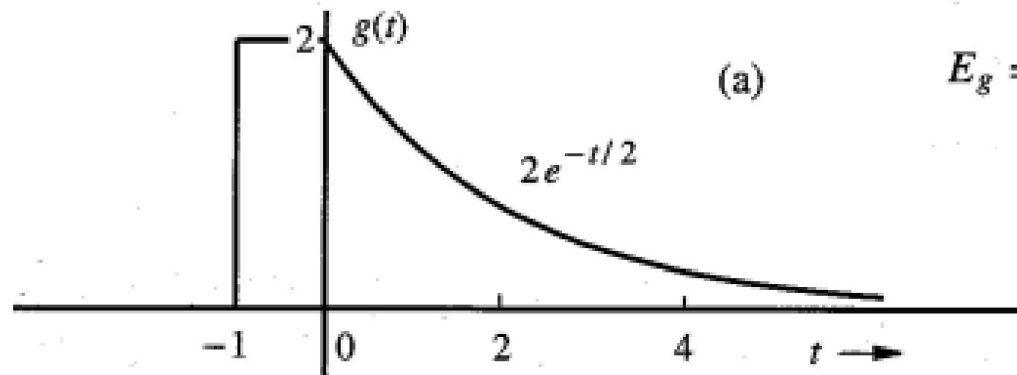
Find P_g or E_g



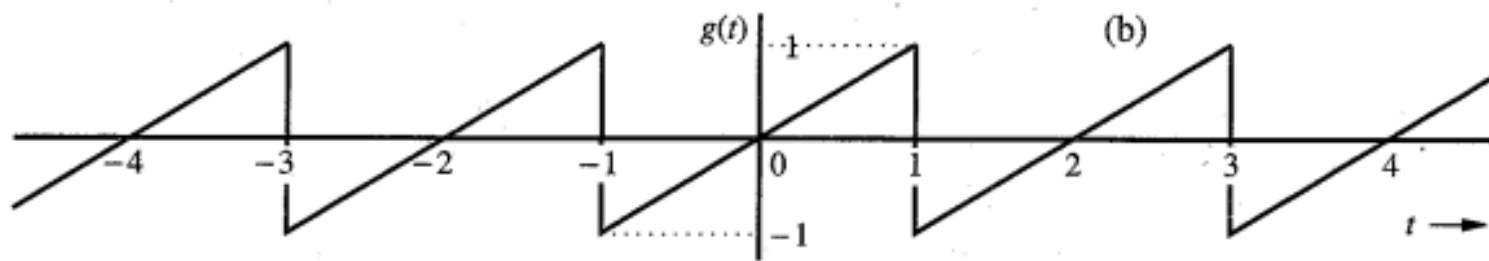
$$E_g = \int_{-\infty}^{\infty} g^2(t) dt = \int_{-1}^0 (2)^2 dt + \int_0^{\infty} 4e^{-t} dt$$
$$= 4 + 4 = 8$$



Find P_g or E_g



$$E_g = \int_{-\infty}^{\infty} g^2(t) dt = \int_{-1}^0 (2)^2 dt + \int_0^{\infty} 4e^{-t} dt$$
$$= 4 + 4 = 8$$



$$P_g = \frac{1}{2} \int_{-1}^1 g^2(t) dt = \frac{1}{2} \int_{-1}^1 t^2 dt = \frac{1}{3}$$